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(The space for each part should be expanded as required, but the report overall should not exceed 12 pages.)

Part I

(a) Electrical parameter extraction (methods and results)

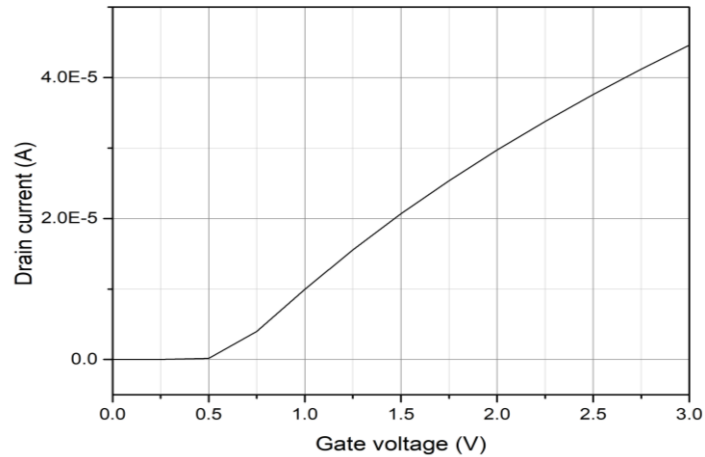


Figure 1 Transfer characteristic

i. I_{on} , when the gate voltage is 3V

I used the HP4145 tool to find it: I rotated the dial to 3 and observed the change in the value of the y-parameter of the graph, therefore

$$I_{on} = 4.459 \times 10^{-05} \text{ A}$$

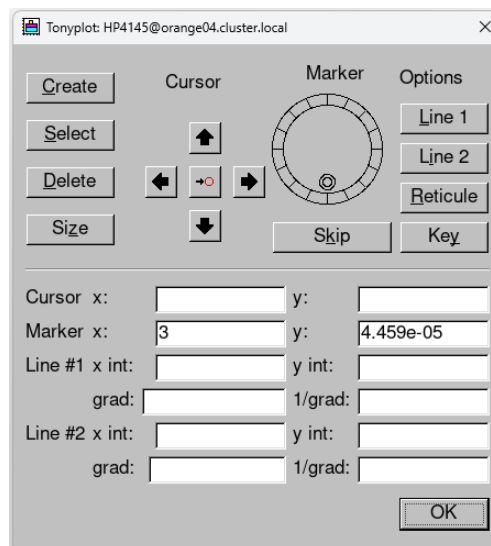


Figure 2 On current using HP14145 tool

ii. Off current

Similarly, when the dial is rotated to 0, we get the off current, therefore,

$$I_{off} = 5.382 \times 10^{-13} \text{ A}$$

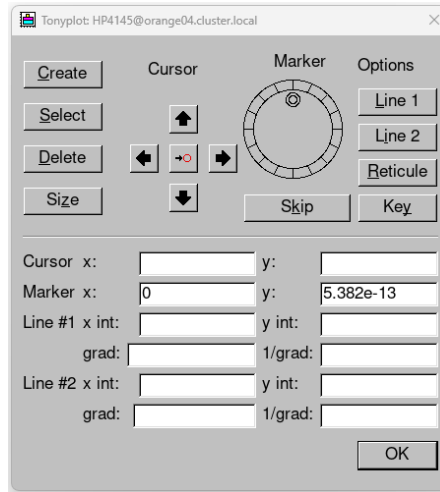


Figure 3 Off current using the HP4145 tool

iii. Threshold voltage

V_{th} can be found directly from the graph by observing the voltage where the current starts to enter the subthreshold region or by extrapolating the values, i.e.

```
EXTRACT> extract name="nvt" (xintercept(maxslope(curve(abs(v."gate"),abs(i."drain")))) - abs(ave(v."drain"))/2.0)
nvt=0.534385821
```

Figure 4 Extracted threshold voltage

V_{th} can also be extracted by taking the maximum slope of the I_d Vs V_g graph, which intersects at the X-axis and subtracting half of the drain voltage from it.

$$V_{th} = 0.534386 \text{ V}$$

iv. Subthreshold Swing

It can be found by taking the inverse slope of the log/linear plot of the transfer characteristics.

$$S = 0.0897 \frac{V}{decade}$$

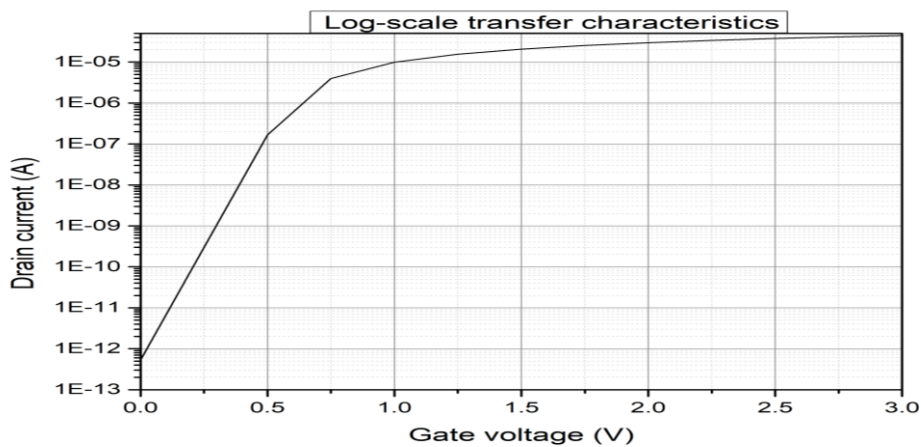


Figure 5 Log-scale transfer characteristic

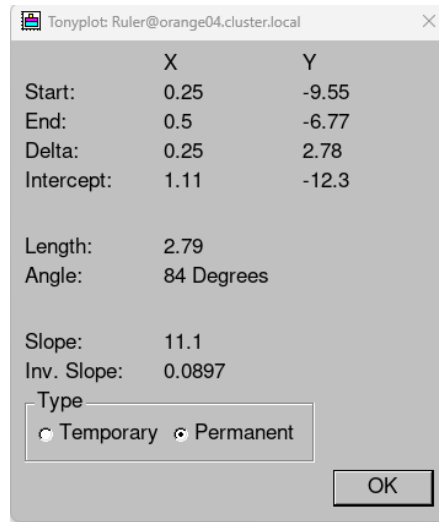


Figure 6 Sub-threshold swing using HP14145 tool

(b) Physical device parameter extraction (methods and results)

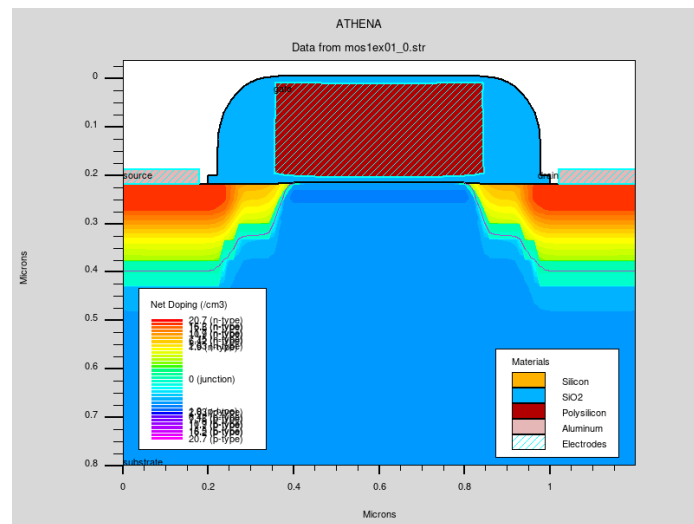


Figure 7 ATHENA Plot of MOSFET

i. Gate oxide thickness

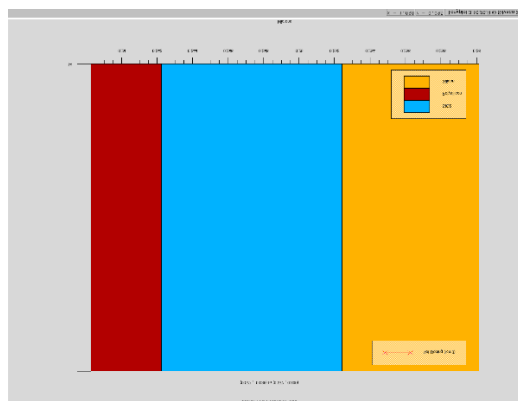


Figure 8 Measure of gate oxide thickness

It is calculated by using the vertical cutline tool. Then a ruler is used to measure the oxide thickness.
 Gate oxide thickness = 0.010 microns (0.242 to 0.252)

$$t_{ox} = 1.0 * 10^{-8} m$$

ii. **Body doping**

It can be calculated by using a horizontal cutline tool on the body part of the ATHENA plot.

$$\text{Body doping} = N_A$$

$$N_A = 10^{17} \text{ cm}^{-3} = 10^{23} \text{ m}^{-3}$$

iii. **Gate material**

Fig. 2 shows the gate material used, i.e. Polysilicon

It can also be extracted using the probe tool and in the output section.

iv. **Gate doping**

It can be calculated by using the horizontal cutline tool at the gate part from source to drain, and then using the display tool to measure Net doping.

$$\text{Gate doping} = 10^{20.3}/\text{cm}^3 \text{ or } 2 \cdot 10^{20}/\text{cm}^3$$

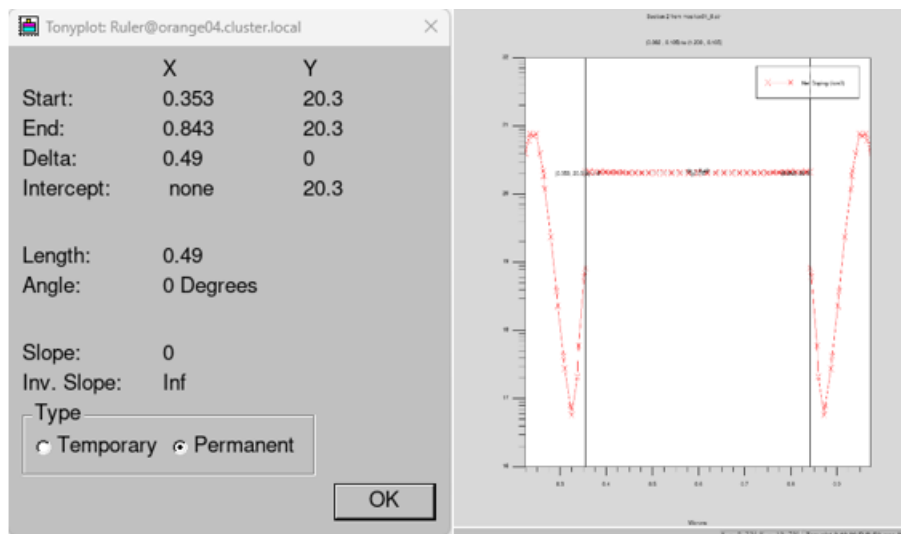


Figure 9 Measure of gate doping using the cutline and ruler tools

v. **Gate length**

It can be calculated by using the horizontal cutline tool and then using a ruler to measure the length.

$$\text{Gate length (L)} = 0.49 \text{ microns (0.353 to 0.843)}$$

(c) **On current with drain voltage at 3V (method and result)**

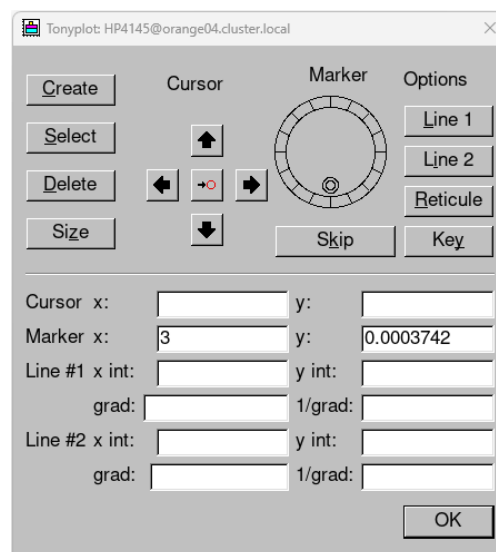


Figure 10 On current at $V_g = 3 \text{ V}$

$$I_{on} = 3.742 * 10^{-4} A$$

(d) Theoretical MOSFET parameters (methods and results)

$$\epsilon_{SiO_2} = 3.45 * 10^{-11} \frac{F}{m} \quad \epsilon_{Si} = 1.04 * 10^{-10} \frac{F}{m} \quad q = 1.602 * 10^{-19} C \quad V_T = \frac{kT}{q} = 0.026$$

$$n_i = 10^{10} cm^{-3} \quad N_A = 10^{17} cm^{-3} = 10^{23} m^{-3} \quad L = 0.4914 * 10^{-6} m \quad W = 1 * 10^{-6} m$$

$$\mu = 0.05 \frac{m^2}{V.s} \quad t_{ox} = 1.01 * 10^{-8} m$$

$$Q_{ox} = q * 10^{10} \quad Q_{ox} = 4.81 * 10^{-5} \frac{C}{m^2}$$

Fermi bulk potential:

$$\phi_F = V_T \ln\left(\frac{N_A}{n_i}\right) \quad \phi_F = 0.026 \ln\left(\frac{10^{17} cm^{-3}}{10^{10} cm^{-3}}\right) \quad \phi_F = 0.4190 V$$

Semiconductor workfunction:

$$\phi_S = \chi + \frac{E_g}{2} + \phi_F \quad \phi_S = 4.05 + \left(\frac{1.12}{2}\right) + 0.4190 \quad \phi_S = 5.029 V$$

Gate metal workfunction:

$$\phi_M = 4.17 V$$

Depletion charge density:

$$Q_d = \sqrt{4qN_A \epsilon_{Si} \phi_F}$$

$$Q_d = \sqrt{4 * 1.609 * 10^{-19} C * 10^{23} m^{-3} * 1.04 * 10^{-10} \frac{F}{m} * 0.4190 V}$$

$$Q_d = 1.67 * 10^{-3} \frac{C}{m^2}$$

Oxide capacitance:

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} \quad C_{ox} = \frac{3.45 * 10^{-11} \frac{F}{m}}{1 * 10^{-8} m} \quad C_{ox} = 3.45 * 10^{-3} \frac{F}{m^2}$$

Flat band voltage:

Work function difference:

$$\phi_{MS} = \phi_M - \phi_S \quad \phi_{MS} = 4.17 - 5.029 \quad \phi_{MS} = -0.859 V$$

$$V_{FB} = \phi_{MS} - \frac{Q_{ox}}{C_{ox}} \quad V_{FB} = -0.859 - \frac{4.81 * 10^{-5} \frac{C}{m^2}}{3.45 * 10^{-3} \frac{F}{m^2}} \quad V_{FB} = -0.8729 V$$

Threshold voltage:

$$V_{th} = V_{FB} + 2\phi_F + \frac{\sqrt{4q\epsilon_{Si}N_A\phi_F}}{C_{ox}} \quad V_{th} = V_{FB} + 2\phi_F + \frac{Q_d}{C_{ox}} \quad V_{th} = -0.8729 + 0.838 + 0.48$$

$$V_{th} = 0.4451 V$$

Subthreshold swing:

Depletion capacitance:

$$C_{dep} = \sqrt{\frac{q\epsilon_{Si}N_A}{2\phi_F}} \quad C_{dep} = \sqrt{\frac{1.602*10^{-19} C * 1.04 * 10^{-10} \frac{F}{m} * 10^{23} m^{-3}}{2*0.4190 V}} \quad C_{dep} = 1.410 * 10^{-3} \frac{F}{m^2}$$

$$S = (\ln 10)V_T \left(1 + \frac{C_{dep}}{C_{ox}}\right)$$

$$S = (\ln 10) * 0.026 * \left(1 + \frac{1.410 * 10^{-3} \frac{F}{m^2}}{3.45 * 10^{-3} \frac{F}{m^2}}\right) \quad S = 0.0842 \frac{V}{decade}$$

For $V_d = 0.1 V$ (Linear region)

$$I_{ds} = \mu C_{ox} \frac{W}{L} (V_g - V_t) V_{ds}$$

$$I_{on} = 8.96 * 10^{-5} A$$

For $V_d = 0.1 V$ (Subthreshold region)

$$I_{off} = \mu C_{ox} \frac{W}{L} (m - 1) V_T^2 \exp\left(\frac{q(V_g - V_{th})}{m k T}\right) \left(1 - \exp\left(-\frac{q V_{ds}}{k T}\right)\right)$$

$$I_{off} = 5.382 * 10^{-13} A$$

For $V_d = 3 V$ (Saturation region):

Maximum depletion capacitance:

$$C_{dm} = \sqrt{\frac{q N_A \epsilon_{Si}}{4\phi_F}} \quad C_{dm} = \sqrt{\frac{1.602 * 10^{-19} C * 10^{23} m^{-3} * 1.04 * 10^{-10} \frac{F}{m}}{4 * 0.4190 V}} \quad C_{dm} = 0.99 * 10^{-3} \frac{F}{m^2}$$

Body coefficient:

$$m = 1 + \frac{C_{dm}}{C_{ox}} \quad m = 1 + \frac{0.99 * 10^{-3} \frac{F}{m^2}}{3.45 * 10^{-3} \frac{F}{m^2}} \quad m = 1.28$$

$$I_{ds} = \mu C_{ox} \frac{W}{L} \left((V_g - V_t) V_{ds} - \frac{m}{2} V_{ds}^2 \right) \quad I_{ds} = \mu C_{ox} \frac{W}{L} \frac{(V_g - V_t)^2}{2m}$$

$$I_{ds} = 8.94 * 10^{-4} A$$

(e) Comparison between calculated and extracted electrical parameters

Table 1 Comparison between calculated and extracted parameters

Parameters	Calculated	Extracted
V_{FB}	$V_{FB} = -0.8729 V$	$V_{FB} = -0.85 V$
V_{th}	$V_{th} = 0.4451 V$	$V_{th} = 0.534386 V$
I_{on} at $V_d = 0.1 V$	$I_{on} = 8.96 * 10^{-5} A$	$I_{on} = 4.459 * 10^{-5} A$
I_{off} at $V_d = 0.1 V$	$I_{off} = 5.382 * 10^{-13} A$	$I_{off} = 5.382 * 10^{-13} A$
I_{on} at $V_d = 3 V$	$I_{ds} = 8.94 * 10^{-4} A$	$I_{ds} = 3.742 * 10^{-4} A$
Subthreshold swing (S)	$S = 0.0842 \frac{V}{decade}$	$S = 0.0897 \frac{V}{decade}$

Observation:

It can be observed from Table 1 that the calculated results vary slightly from the extracted results that we have obtained from the Silvaco simulation. The reasons for these differences can be:

- The analytical calculations usually use formulas based on the generalized long channel MOSFET equations, while Silvaco uses real physics, mesh profiles, doping profiles, geometry, and many other numerical models, so they slightly vary from the calculated results [1].
- It can also depend on the different parameter values that have been used in the calculations. As I have used a mobility value of $0.05 m^2/Vs$, the simulation may use a different value.

Part II ELEC6256

(a) Plot of electronic band structure and explanation

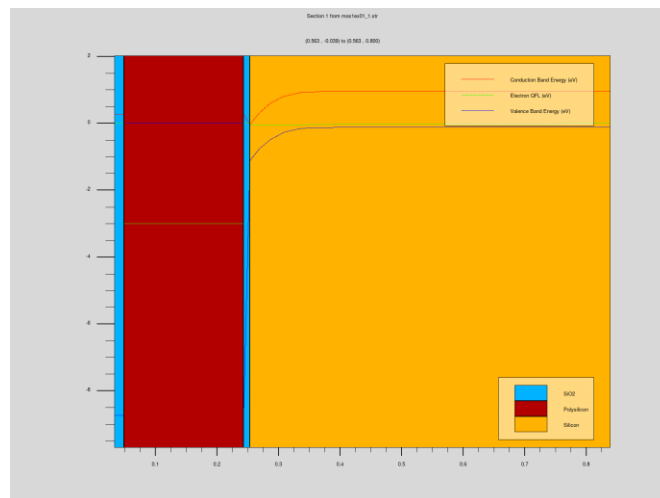


Figure 11 Band bending at the Si-SiO₂ interface at $V_g = 0 V$

The script has been altered according to helpful hint 2, and by using helpful hint 3, the conduction band, valence band, and the Fermi level (Electron QFL) of the MOSFET have been plotted using the cutline tool.

Explanation:

- i. Band bending can occur at the interface of polysilicon, oxide layer, and semiconductor because of the difference in the work function of these layers. When these materials come into contact, their Fermi level must align at equilibrium to find a stable state, so electrons from the layer of the higher energy level flow to the lower energy level. This transfer of electrons causes electric fields, and band bending occurs even when no gate voltage is applied [2] [3].
- ii. The polysilicon gate region is made up of small grains of silicon and is heavily doped to make it conductive. This alteration makes the simulation treat the polysilicon region as a metal-like conductor. Because of high doping, the Fermi level may lie inside the conduction band (for n^+ poly) or inside the valence band (for p^+ poly), which makes the band bending in this region negligible [2] [3].

(b) Plot the electronic band structure in flat band conditions and give the value of V_{FB} .

Flat band can be measured by taking the difference of the work functions of the metal and semiconductor layers, and the sheet charge voltage between the gate and oxide layers.

In Tonyplot, the flat band voltage is the gate voltage at which the energy bands in the semiconductor substrate are flat, which means that the band bending disappears.

Therefore, $V_{FB} = -0.85$ V (When workfunction = 4.17 was given as n.poly)

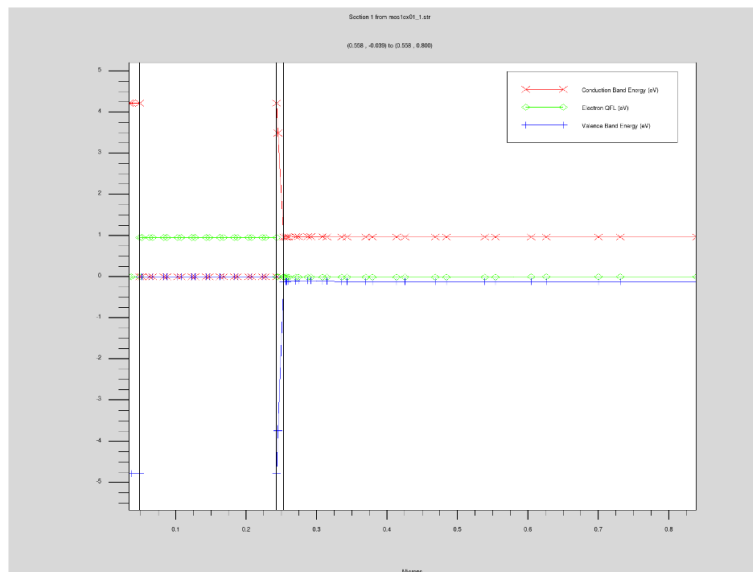


Figure 12 Flat band voltage in simulated MOSFET

(c) Plot V_{FB} vs. gate workfunction and explain intercept values.

Workfunction is the measure of the difference between the reference level (in most cases, i.e. Vacuum) and the Fermi level of a material [2] [3].

In Silvaco/Tcad, work function is an input value, so it must be put manually in the code. After that, we have to ramp the gate voltage in both the positive and negative ranges. In general, the V_{FB} is less than the V_{th} .

Table 2 Relation between work function and flat band voltage

Work function	V_{FB}
3.8	-1.2
4.14	-1.14
4.17	-0.85
4.38	-0.75
4.8	-0.35
5.1	-0.1
5.65	0.2

Observation:

$$V_{FB} = \phi_{MS} - \frac{Q_{ox}}{C_{ox}}$$

The work function of the MOSFET is an input value which is dependent on the gate material used [4].

As the gate material used in this device is Polysilicon, I have scaled the work function in that required range.

The V_{FB} increases with an increase in the gate material work function value, assuming all the other parameters remain constant.

It is evident from Table 2 that as we increase the work function value, the required flat band voltage also increases.

For $\phi_M < \phi_S$ the flat band voltage is negative, which means that the bands are bent upward strongly even at 0 V bias. As the value of the gate work function increases $\phi_M > \phi_S$ the flat band voltage becomes positive. This means that the semiconductor bands bend downward even at zero bias [2] [3] [4].

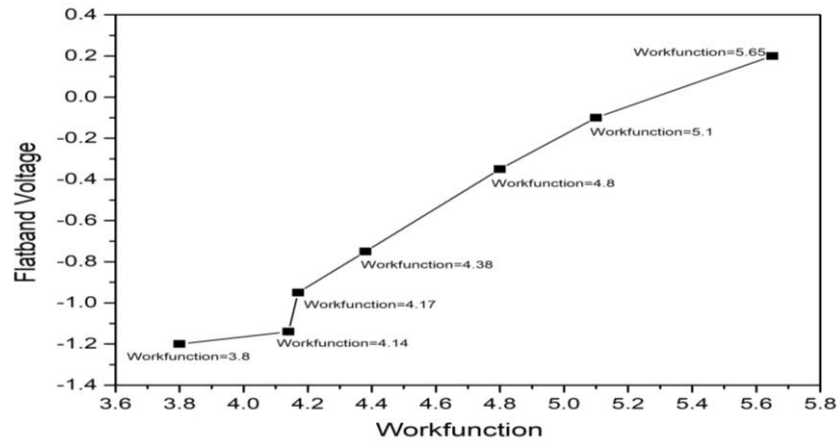


Figure 13 Comparison between work function and Flat band voltage

Since the semiconductor material used is Silicon and the work function value is fixed, Figure 6 shows an almost linear trend [4]. Table 2 and Figure 6 quite clearly show this trend. Based on the graph, this MOSFET is n+ poly n-Si.

Part III

(method, results and discussion from device scaling investigation)

Method:

Table 3 Effects of scaling gate length on leakage current

Cd (gate length)	Leakage current (A)	On current (A)
0.6	2.109e-12	3.746e-05
0.5	2.464e-12	4.309e-05
0.3	6.01e-12	5.324e-05
0.1	6.411e-09	6.975e-05
0.05	0.002608	0.002608

Observation:

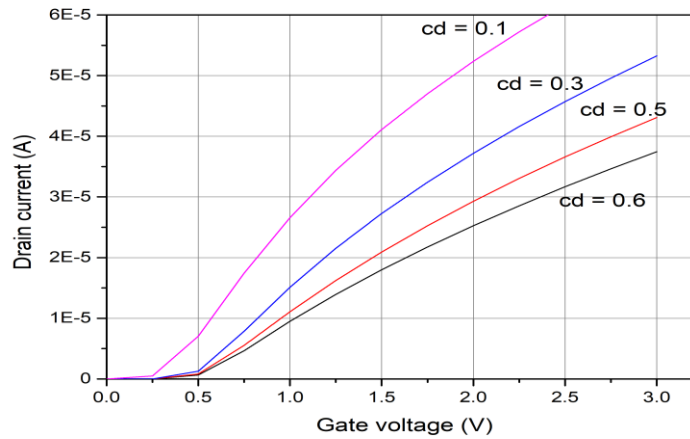


Figure 14 Effect of gate length scaling

Decreasing the gate length while keeping all the parameters constant increases the leakage current [5] [2].

The gate length at which the leakage current becomes too high is $cd = 0.05$. At this length, it has been made clear that the resulting graph reaches towards a linear output plot, i.e. the I_{ds} vs V_{gs} graph gives a linear output.

Result:

Table 4 Initial MOSFET characteristics

Cd	Vth	Leakage current	Ion/Ioff	Subthreshold swing	t_{ox} (angstrom)
0.6	0.458721 V	2.11e-12 A	1.78e+7	0.0843 V/decade	99.8207

Table 5 Scaling MOSFET parameters

Scaling parameter	Value	Vth	Leakage current	Ion	t_{ox} (angstrom)	Subthreshold swing
P-well Boron implant	8e-12	0.450367	2.464e-12	4.309e-05	99.9663	0.0836
	9.5e-12	0.47538	1.79e-12	4.245e-05		
	1e13	0.466694	2.031e-12	4.29e-05		
Gate oxide	Time=10.7sec, Temp=925	0.473065	2.031e-12	4.29e-05	98.0959	0.08

	Time=10.5sec, Temp=925	0.46	2.346e-12	4.333e-05	96.908	0.084
Vth adjust implant	9.5e11 1e12	0.466 0.487502	2.346e-12 1.168e-12	4.333e-05 4.274e-05	96.908	0.0844
Implant phosphorous dose	5.03e13 7e13 1e14	0.471316 0.462163 0.453617	1.872e-12 2.042e-12 2.231e-12	4.71e-05 4.971e-05 5.238e-05	96.908	0.0841 0.0821
Implant Arsenic dose	5e16	0.452957	2.26e-12	5.33e-05	96.908	0.0817
Vd	0.4 0.3	0.432963 0.449238	3.03e-12 2.811e-12	0.0001851 0.0001469	96.908	0.082

Table 6 Final MOSFET parameter values

Cd	P-well implant	Vth adjust implant	Implant Phosphorous dose	Implant Arsenic dose	Vd
0.5	1e13	1e12	1e14	5e16	0.4

Table 7 Improved MOSFET characteristics

Ion/Ioff	Vth	Subthreshold swing	tox (angstrom)
6.108e+7	0.432963 V	0.0817 V/decade	96.908

Discussion:

The following are the parameters that can be scaled to improve the MOSFET characteristics:

- **Gate length (L):** Reducing L increases the switching speed by reducing gate delay, but ultra-short L can cause V_{th} roll off, DIBL and can increase leakage current.
It can be scaled by changing “cd” values.
- **Gate oxide thickness (t_{ox}):** A thin t_{ox} increases C_{ox} , I_{on} and switching speed, but if it is too thin then it causes gate leakage due to tunnelling.
It is dependent on the temperature and the diffusion time required for the process. So, to deposit a thin oxide layer, we have to decrease the diffusion time and scale the temperature accordingly.
- **Vth:** It is ideal to have V_{th} as low as possible to make the switching speed higher, but it also results in higher leakage current. On the other hand, if it's too high, then the device becomes slow.
 V_{th} can be decreased by increasing the Boron implant doping, as then it will require less free charge to form a conduction channel.
- **Phosphorous and Arsenic doping:** These are n-type dopants that are used to improve I_{on} , but if too high, the leakage current also increases.

From Table 5, it is evident that certain characteristics of the MOSFET change. The most noticeable changes of an improved MOSFET device are:

- **Reduction in leakage current (I_{off}):** This means that there is lower static power consumption, less device heating, better battery life, and reduced standby power. It improves energy efficiency and reliability.
- **Increase in I_{on} :** It indicates faster switching and higher transconductance gain.
- **Reduction in threshold voltage (V_{th}):** It means that the device requires less power to function, but too low and it increases leakage current.
- **Increase in the I_{on}/I_{off} ratio:** This means that the MOSFET now requires less power for switching and has overall better control of the current flow.

The evidence on these improvements can also be found in [3] [5] [6].

Tables 4, 6, and 7 show the stark differences between the initial and the final MOSFET characteristics. The improved characteristics are based on the 2000s ITRS Roadmap [7], and all the extracted values from scaling the MOSFET parameters complement the results that were expected in the report [7].

References:

- [1] L. T. Pang and B. Nikolić, "Measurements and analysis of process variability in 90 nm CMOS," *IEEE J. Solid-State Circuits*, vol. 44, no. 5, pp. 1655–1663, 2009, doi: 10.1109/JSSC.2009.2015789.
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- [4] E. Erben, K. Hempel, and D. Triyoso, "Work Function Setting in High-k Metal Gate Devices," *Complement. Met. Oxide Semicond.*, 2018, doi: 10.5772/intechopen.78335.
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- [7] J. Bruines *et al.*, "2001 ITRS Executive Summary," p. 68, 2001.